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Potentially Inappropriate Medication Use among Patients with Dementia in Traditional Medicare and Medicare Advantage

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Abstract

Background: Older adults with dementia are susceptible to receiving potentially inappropriate medications (PIMs), where the risks likely outweigh the benefits. Medicare Advantage prescription drug plans (MA-PDs) cover both medical and prescription drug benefits, creating a financial incentive to reduce PIM use and unnecessary healthcare costs from adverse drug events, whereas standalone Medicare prescription drug plans (PDPs) used by Traditional Medicare beneficiaries are only responsible for outpatient prescription drug costs.

Objective: Compare the use of PIMs between PDP and MA-PD enrollees with dementia.

Methods: Using 2016-2019 Medicare claims and encounter data, we estimated the associations between Medicare enrollment type and PIM use: 1) potentially harmful drug-disease interactions in older adults with dementia; 2) potentially harmful drug-disease interactions in older adults with dementia and a history of falls; and 3) high-risk medication use in older adults.

Results: MA-PD enrollees had significantly lower utilization of PIMs than standalone PDP enrollees: a 0.7 percentage-point [95% CI: 0.5, 0.8] lower prevalence of potentially harmful drug-disease interactions in older adults with dementia; a 3.1 percentage-point [2.6, 3.5] lower prevalence of potentially harmful drug-disease interactions in older adults with dementia and a history of falls; and a 0.5 percentage-point [0.4, 0.6] lower prevalence of high-risk medications in older adults.

Conclusions: MA-PD enrollees with dementia experienced consistently lower prevalence of PIM use than those in PDP. As Medicare Advantage enrollment continues to grow, it will be increasingly important to identify and leverage the features of MA-PD plans that promote safe medication prescribing for Medicare beneficiaries with dementia.

Keywords

dementia; potentially harmful prescription drugs; Medicare

INTRODUCTION

Alzheimer's disease and related dementias (ADRD) can lead to a progressive decline in cognitive functioning that impairs memory, concentration, language, and decision-making.¹ In 2023, approximately 6.7 million Americans age 65 and older had ADRD, and this number is projected to grow to 14 million by 2060.¹ A majority have multiple comorbid chronic conditions that require continuous disease management through prescription medications.¹⁻³ Because of their complex care needs, older adults with ADRD are susceptible to receiving potentially inappropriate medications (PIMs), where the risks likely outweigh the benefits.⁴ Patients receiving PIMs are more likely to experience medication side effects, falls, hospitalizations, and increased healthcare costs.⁴

Most older adults with ADRD are enrolled in Medicare.¹ Medicare beneficiaries can choose to receive coverage through Traditional Medicare (TM) and optionally add Part D prescription drug coverage through standalone prescription drug plans (PDPs). Alternatively, Medicare beneficiaries can enroll in Medicare Advantage (MA) plans with Part D prescription drug benefits (MA-PDs) that cover medical services and prescription drugs under 1 plan like other commercial plans. Enrollment in MA-PDs has been steadily growing and reached 56% of Medicare beneficiaries with Part D coverage in 2023.⁵

MA-PD's capitated payment system and financial alignment of medical and prescription drug services create strong incentives for plans to prevent unnecessary healthcare spending due to adverse drug events.^{4,6,7} This is especially true for enrollees with ADRD, who have substantially higher costs of care and are more susceptible to adverse drug events.^{1,4,8} These financial incentives may motivate MA-PD plans to selectively contract with higher quality providers who prescribe fewer PIMs and to implement more robust Medication Therapy Management (MTM) programs that can decrease PIM utilization.^{6,9,10} In contrast, standalone PDPs cover only prescription drugs and are not financially responsible for the medical consequences of adverse drug events from PIMs.

Despite the stronger inherent financial incentives, it is unclear whether MA-PDs reduce PIM utilization among enrollees with ADRD, compared with standalone PDPs. For instance, MA-PD plans often reduce enrollees' out-of-pocket drug costs.¹¹ While this may promote medication adherence¹² and decrease medical spending for poorly controlled chronic conditions,¹³ it could unintentionally increase exposure to PIMs. Given the vulnerability of Medicare beneficiaries with ADRD and their high medical spending, it is important to understand how PIM use differs by enrollment types. Therefore, this study compared the use of PIMs between PDP and MA-PD enrollees with ADRD.

METHODS

Data

This pooled cross-sectional study used 2016-2019 TM claims and MA encounter data. The 2016 data were used only for the reference period to identify the study population; outcomes were analyzed for 2017-2019. We used the Part D Event File, which provides outpatient prescription drug claims for both PDP and MA-PD enrollees, to construct measures of PIM utilization. This file includes details on drug name, strength, days supply, and prescription fill date. We used the Master Beneficiary Summary File for beneficiary demographics, ZIP code and county of residence, and monthly Medicare and Medicaid enrollment details, including plan type. We used the Medicare carrier, inpatient, and outpatient files to identify diagnoses and primary care providers. We also linked county healthcare resources from the Area Health Resource File and ZIP code socioeconomic characteristics from the American Community Survey.

Study Population

We identified Medicare beneficiaries age 65 and older with ADRD **during the outcome measurement year or prior year**, based on algorithms from the Centers for Medicare & Medicaid Services (CMS) Chronic Condition Warehouse.¹⁴ The study includes beneficiaries with continuous Medicare Part A, B, and D coverage for 12 months in a specific year, excluding those who switched between TM and MA midyear. Due to concerns of missing records in MA encounter data, we followed established and validated methods to limit analysis to plans with highly complete data.^{15,16} We excluded beneficiaries with Medicare entitlement solely due to end-stage renal disease or living outside the 50 U.S. states or Washington, D.C. To directly compare PDP versus MA-PD outcomes, we excluded beneficiaries enrolled in Medicare plans other than TM and MA, or that allow PDP enrollment separately from a MA plan.

Outcomes

We measured 3 PIM outcomes, following definitions from the Healthcare Effectiveness Data and Information Set (HEDIS) quality measures.^{17,18} The first outcome, potentially harmful drug-disease interactions in older adults with dementia, measures the proportion of Medicare beneficiaries with ADRD who filled a prescription for a potentially harmful medication. The second outcome is potentially harmful drug-disease interactions in older adults with dementia who had an accidental fall or hip fracture during the same calendar year. The third outcome, use of high-risk medications in older adults, is the proportion of Medicare

beneficiaries who were dispensed the same high-risk medication at least twice during the calendar year.¹⁹ Each outcome is a period prevalence measured as the proportion of the sample with the outcome in a calendar year. The list of drug classes and excluded conditions for each outcome are detailed in the Outcome Measures section of the Supplemental Digital Content. Different eligibility criteria for each outcome resulted in separate analytic samples.

Statistical Analysis

We estimated linear probability models for the associations between PDP versus MA-PD enrollment and each outcome. Following prior literature,²⁰⁻²² we used linear probability models given the computational feasibility with multiple high-dimensional fixed effects and the ease of interpretation with direct marginal effects.²³ We used inverse probability of treatment weighting (IPTW) to balance observed characteristics between enrollees in PDP and MA-PD for each outcome. We ensured that the covariates were balanced by comparing the standardized mean difference before and after IPTW (Supplemental Digital Content, Tables S1-S3).²⁴ We limited analysis to beneficiaries for whom the propensity score distributions of PDP and MA-PD enrollees overlapped.²⁵ Because individuals could be observed multiple times during the study period, we used robust standard errors clustered by individuals.

Control variables included age, sex, race, ethnicity, Medicaid dual-eligibility, a frailty score,²⁶ and health status measured by Hierarchical Condition Category risk score that omitted health risk assessment and chart review records due to concerns of higher intensity coding for MA enrollees.²⁷ ZIP code-level characteristics included rurality, percent of adults with a 4-year degree, median household income, and percent of population speaking only English. Time-varying county health resources included densities of hospital beds and physicians. To focus on differences between PDP and MA-PD, we also considered the influence of primary care providers who see patients in both plan types. We used provider practice fixed effects to control for idiosyncratic prescribing and care management patterns. Beneficiaries were attributed to a primary care practice, identified by tax identification numbers, based on the majority of outpatient medical claims associated with that practice during a calendar year. Beneficiaries not attributed to a primary care practice were attributed based on claims from all provider specialties. MA plan competition in different markets that may lead to changes in provider practices was controlled by the county-level MA Herfindahl-Hirschman Index. County fixed effects controlled for time-invariant county characteristics, and year fixed effects accounted for historical trends.

RESULTS

As shown in Table 1, the study included 3,363,797 beneficiary-year observations (64.7% PDP, 35.3% MA-PD) eligible for potentially harmful drug-disease interactions with dementia, 464,254 beneficiary-year observations (66.1% PDP, 33.9% MA-PD) eligible for potentially harmful drug-disease interactions with dementia and a history of falls, and 4,610,658 beneficiary-year observations (67.0% PDP, 33% MA-PD) eligible for high-risk medication use in older adults.

Table 2 shows that the unadjusted prevalence of potentially harmful drug-disease interactions with dementia was 33.6% for PDP enrollees and 30.7% for MA-PD enrollees. The unadjusted prevalence of potentially harmful drug-disease interactions with dementia and a history of falls was 38.3% for PDP enrollees and 33.4% for MA-PD enrollees. The unadjusted prevalence of high-risk medication use was 16.4% for PDP enrollees and 14.2% for MA-PD enrollees.

The regression-adjusted prevalence (Table 2) of potentially harmful drug-disease interactions with dementia was 32.9% [95% CI: 32.8%, 33.0%] for PDP enrollees and 32.2% [32.1%, 32.3%] for MA-PD enrollees. The adjusted prevalence of potentially harmful drug-disease interactions with dementia and a history of falls was 37.6% [37.4%, 37.9%] for PDP enrollees and 34.5% [34.2%, 34.9%] for MA-PD enrollees. The adjusted prevalence of high-risk medication use was 16.0% [16.0%, 16.1%] for PDP enrollees and 15.5% [15.4%, 15.6%] for MA-PD enrollees.

Regression results (Table 2) indicate that MA-PD enrollees had significantly lower PIM utilization than PDP enrollees, including a 0.7 percentage-point [0.5, 0.8] lower prevalence of potentially harmful drug-disease interactions with dementia, a 3.1 percentage-point [2.6, 3.5] lower prevalence of potentially harmful drug-disease interactions with dementia and a history of falls, and a 0.5 percentage-point [0.4, 0.6] lower prevalence of high-risk medications. Full regression results are presented in Table S4 of the Supplemental Digital Content.

DISCUSSION

Individuals with ADRD are prone to the consequences of drug interactions as a result of inappropriately prescribed medications. In our study of older Medicare beneficiaries with ADRD, PIM utilization was highly prevalent. Approximately one-third of the eligible study population received medications that can cause harmful interactions with ADRD or a history of falls; nearly one-sixth received medications that are likely to cause harm to older adults. Moreover, we found that those with ADRD who enrolled in MA-PDs had consistently lower PIM utilization, compared to PDP enrollees.

Our findings are consistent with previous studies that found lower rates of high-risk medication use among MA enrollees in the general Medicare population.²⁸ However, our findings indicate that Medicare beneficiaries with ADRD are more greatly affected by PIMs. We observed prevalence of PIM utilization among individuals with ADRD 3 to 4 times higher than the general Medicare population.²⁸ This suggests that the quality of medication-related outcomes should be closely monitored for individuals with ADRD and other groups who are particularly vulnerable to receiving PIMs and experiencing subsequent adverse drug outcomes.

There are several possible explanations for the lower prevalence of PIM utilization among MA-PD enrollees with ADRD. First, MA-PD plans may implement stronger utilization management tactics to reduce PIM use for enrollees with ADRD, who are at high risk of adverse drug events. Second, MA-PD plans may promote safe medication use by contracting

with networks of high quality providers who avoid prescribing PIMs to older adults with ADRD.⁹ Last, MA-PD enrollees have higher rates of comprehensive medication reviews as part of MTM programs, some of which target patients with ADRD,²⁹ which may lead to lower PIM utilization.³⁰

Our findings have important policy implications. The consistently lower prevalence of PIM utilization among MA-PD enrollees with ADRD suggests that CMS should leverage MA-PD's integration of medical and prescription drug benefits by incentivizing safe medication management practices in this vulnerable population. While there are currently several Part D 'display measures' pertaining to safe medication use,³¹ none factor into Part D plans' star ratings for either PDP or MA-PD. CMS could apply existing measures of PIM use in persons with dementia to count toward plans' star ratings. Additionally, CMS recently finalized rule changes to Part D MTM programs **requiring that** plans include ADRD as a targeted core condition.³² However, plans can require that enrollees take up to 8 chronic medications to be eligible for MTM. Reducing the required number of medications would allow more beneficiaries with ADRD to benefit from comprehensive medication reviews to reduce PIM utilization, particularly for PDP enrollees who typically face more restrictive MTM eligibility criteria.³³

Limitations

This study has several limitations. First, the study did not attempt to infer a causal relation. While we used IPTW to address selection bias, unobserved patient characteristics that **impact** enrollment choices may remain as biases.³⁴ Second, the latest data examined in this study are from 2019. Although more recent data are available from CMS, the newer data largely cover the COVID-19 pandemic, which interrupted healthcare use patterns in the U.S. Third, although the HEDIS algorithms are validated standards, prescription drug claims may overestimate PIM exposure for patients that fill prescriptions but do not take them.

CONCLUSIONS

Medicare beneficiaries with ADRD who enrolled in MA-PD plans experienced consistently lower PIM utilization, compared to PDP enrollees. As MA-PD enrollment continues to grow, it will be increasingly important to identify and leverage the features of MA-PD plans that promote safe medication prescribing for Medicare beneficiaries with ADRD.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Table 1. Characteristics of Medicare beneficiaries with Alzheimer's disease and related dementias, by enrollment, 2017-2019

	Potentially harmful drug-disease interaction – dementia		Potentially harmful drug-disease interaction – dementia and history of falls		High-risk medications in older adults	
	PDP	MA-PD	PDP	MA-PD	PDP	MA-PD
Number of patient-year observations	2,176,149	1,187,648	307,104	157,150	3,087,835	1,522,823
Number of patients	1,178,528	715,036	227,094	122,578	1,563,273	886,684
Age, mean (SD)	83.9 (7.5)	83.0 (7.3)	86.3 (7.0)	85.7 (6.9)	83.0 (7.9)	82.4 (7.5)
Female, %	68.1	64.7	70.1	66.5	67.9	64.8
American Indian or Alaskan Native (non-Hispanic), %	0.4	0.1	0.4	0.2	0.4	0.2
Asian (non-Hispanic), %	3.4	2.5	3.3	2.4	3.1	2.4
Black (non-Hispanic), %	9.2	13.8	7.7	11.8	10.1	14.0
Hispanic (any race), %	6.5	11.6	5.4	8.6	6.7	11.5
White (non-Hispanic), %	79.8	71.3	82.5	76.4	79.1	71.4
Other race, %	0.7	0.6	0.6	0.6	0.7	0.6
Rural residence, %	25.5	15.3	25.9	16.0	24.6	15.2
Dual-eligible, %	37.6	26.3	32.2	23.8	43.7	28.9
HCC risk score, mean (SD)	1.85 (1.38)	1.70 (1.25)	1.85 (1.26)	1.73 (1.18)	2.03 (1.52)	1.78 (1.33)
Kim frailty score, mean (SD)	0.23 (0.08)	0.21 (0.08)	0.23 (0.08)	0.23 (0.08)	0.23 (0.08)	0.22 (0.08)
ZIP code median household income, mean (SD)	65646 (27843)	62695 (24626)	66536 (28444)	64023 (25231)	65408 (27855)	62723 (24726)
Percentage of ZIP code with a 4-year degree, mean (SD)	31.7 (17.2)	30.1 (15.9)	32.4 (17.6)	31.1 (16.3)	31.7 (17.2)	30.2 (16.0)
Percentage of ZIP code households speaking only English, mean (SD)	82.1 (19.0)	79.7 (20.9)	82.9 (18.0)	81.4 (19.0)	81.7 (19.2)	79.7 (20.8)
Hospital beds per 1000 population (county level), mean (SD)	3.03 (2.24)	3.07 (1.98)	3.05 (2.23)	3.08 (2.10)	3.05 (2.23)	3.08 (1.99)
Physicians per 1000 population (county level), mean (SD)	3.31 (2.45)	3.53 (2.35)	3.36 (2.50)	3.57 (2.45)	3.35 (2.48)	3.55 (2.37)
Medicare Advantage county HHI, mean (SD)	2600 (1527)	2267 (1108)	2622 (1537)	2325 (1139)	2558 (1491)	2259 (1101)

Notes: % presents percent of patient-year observations within a column.

Abbreviations: PDP, Medicare Part D standalone prescription drug plan; MA-PD, Medicare Advantage plan with prescription drug benefits; HCC, Hierarchical Condition Category; HHI, Herfindahl-Hirschman Index

Table 2. Differences in potentially inappropriate medication use between PDP and MA-PD enrollees with Alzheimer’s disease and related dementias

Outcome	Unadjusted Prevalence, %		Regression-adjusted Prevalence, % (95% CI)		Average Marginal Effects, percentage points (95% CI)
	PDP	MA-PD	PDP	MA-PD	
Potentially harmful drug-disease interaction with ADRD	33.6	30.7	32.9 (32.8, 33.0)	32.2 (32.1, 32.3)	-0.7 (-0.8, -0.5)
Potentially harmful drug-disease interaction with ADRD and history of falls	38.3	33.4	37.6 (37.4, 37.9)	34.5 (34.2, 34.9)	-3.1 (-3.5, -2.6)
High-risk medications in older adults	16.4	14.2	16.0 (16.0, 16.1)	15.5 (15.4, 15.6)	-0.5 (-0.6, -0.4)

Notes: Unadjusted **prevalence** represents the percent of each enrollment group experiencing the outcome. Regression-adjusted **prevalence** is the predicted percent of each enrollment group experiencing the outcome, based on regression models adjusting for individual characteristics, ZIP code socioeconomic characteristics, county health resource availability, and fixed effects for year, county, and primary care practice. Average marginal effects represent the absolute difference in outcomes for MA-PD enrollees compared to PDP enrollees.

Abbreviations: ADRD, Alzheimer’s disease and related dementias; PDP, Medicare Part D standalone prescription drug plan; MA-PD, Medicare Advantage plan with prescription drug benefits